ABOLFAZL RASTGAR MOGHADAM

M.Sc. in Computer Engineering - Computer Networks | Academic CV

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ACADEMIC PROFILE

M.Sc. graduate in Computer Engineering, specializing in Computer Networks, from Iran University of Science and Technology (IUST). My research direction connects applied machine learning, network and cybersecurity, smart-grid communication security, and practical systems implementation. My graduate work focused on AI-based intrusion detection for smart-grid communication environments, while my professional experience in data engineering and observability supports reproducible experimentation, reliable pipelines, and implementation-oriented research in networked systems. I am seeking graduate research opportunities that combine AI-driven security analysis, network security, cyber-physical systems, smart-grid cybersecurity, and related data-intensive infrastructure.

RESEARCH DIRECTION

- AI for security: intrusion detection, anomaly detection, attack classification, feature engineering, model evaluation, and threshold-based decision logic.
- Network and cyber-physical security: TCP/IP-based smart-grid communication, protocol-aware threat analysis, malicious-traffic behavior, and critical-infrastructure protection.
- Implementation-aware systems research: reliable telemetry/data pipelines, observability, automation, and database-backed monitoring for operational environments.

RESEARCH INTERESTS

AI-assisted intrusion detection and anomaly detection in networked systems	Network security, cybersecurity, malicious-traffic analysis, and threat modeling
Smart-grid cybersecurity, IEC-104 traffic analysis, and critical infrastructure protection	Machine learning for security monitoring and unseen-attack awareness
Distributed data systems, observability, and database-supported security monitoring	Implementation-aware systems for reliable security monitoring

EDUCATION

Master of Science in Computer Engineering - Computer Networks

Iran University of Science and Technology (IUST), Tehran, Iran | 2020-2023

- GPA: 16.96/20 (unofficial estimated equivalent: approximately 3.39/4.0)
- National M.Sc. entrance examination: Ranked 178 among approximately 25,000 candidates
- Master’s thesis: AI-Based Intrusion Detection in Smart Grids
- Master’s seminar: Security Challenges of TCP/IP-Based Smart Grids
- Selected coursework: Advanced Computer Networks, Advanced Network Security, Network Management, Wireless and Mobile Networks, Distributed Systems, Operating Systems, Performance Evaluation of Computer Systems

Bachelor of Science in Computer Engineering

Sadjad University, Mashhad, Iran | Graduated 2017

- Bachelor’s thesis: Shortest Path Search Using Ant Colony Optimization (ACO)
- Selected coursework: Algorithms, Data Structures, Discrete Mathematics, Computer Architecture, Programming, Database Systems

RESEARCH COMPETENCIES

- Security analysis: threat modeling, attack classification, IDS/NIDS concepts, network-security reasoning, and critical-infrastructure security requirements.
- Applied ML: preprocessing, feature engineering, supervised classification, anomaly-aware reasoning, cross-validation, threshold calibration, and metric-based evaluation.
- Networked systems: TCP/IP, distributed systems, operating systems, monitoring, reliability, and performance-oriented troubleshooting.
- Research execution: literature review, problem framing, Python-based experimentation, structured evaluation, and concise academic reporting.

THESIS AND ACADEMIC RESEARCH

Master's Thesis: AI-Based Intrusion Detection in Smart Grids | IUST, Tehran, Iran | 2020-2023

Research question: How can ML-based IDS detect malicious behavior in smart-grid communication while considering availability, integrity, and operational reliability?

- Framed smart-grid IDS as a network-security/CPS problem and built a Python-based ML workflow for preprocessing, feature engineering, ensemble classification, cross-validation, and metric-based evaluation.

Master's Seminar: Security Challenges of TCP/IP-Based Smart Grids | IUST, Tehran, Iran | 2020-2022

Research question: What TCP/IP-related security risks affect smart-grid reliability and protection requirements?

- Reviewed core security requirements and mapped communication-layer attacks to smart-grid reliability risks.

Bachelor's Thesis: Shortest Path Search Using Ant Colony Optimization | Sadjad University, Mashhad, Iran | 2014-2017

- Implemented an ACO-based shortest-path method in MATLAB.

MANUSCRIPT IN PREPARATION

Unseen-Attack-Aware Intrusion Detection for Smart-Grid IEC-104: A Four-Stage Hierarchical and Edge-Efficient HIDS

Manuscript in preparation - journal selection stage; approved by supervisor for journal targeting

Preparing a follow-on manuscript on hierarchical IDS for IEC-104 smart-grid traffic. The work aims to separate normal/malicious behavior, classify known attacks, and improve awareness of unseen attack patterns in edge-constrained substation environments.

- Develops a staged IDS pipeline for normal/attack separation, DoS categorization, known-attack classification, and unseen-attack awareness.
- Combines supervised classification with anomaly-detection logic and confidence/threshold-based routing for uncertain windows.
- Uses Python-based experimentation for feature processing, model training, threshold calibration, and standard classification-metric evaluation.

TECHNICAL SKILLS

Programming and experimentation	Python, SQL, T-SQL, Bash, MATLAB, JavaScript
Machine learning and data analysis	Scikit-Learn, Pandas, NumPy, Random Forest, feature engineering, cross-validation, model evaluation
Security and intrusion detection	Network security, IDS/NIDS, anomaly detection, threat modeling, attack classification, malicious-traffic analysis
Networks and systems	TCP/IP, computer networks, network management, distributed systems, operating systems, performance evaluation
Smart-grid and CPS security	Smart-grid security, IEC-104-related IDS research, critical infrastructure protection, operational reliability
Data and distributed infrastructure	SQL Server, PostgreSQL, ClickHouse, Elasticsearch, Redis, Kafka, Debezium, Spark, ETL/ELT, CDC
Observability and tools	Grafana, Prometheus, Zabbix, Redgate SQL Monitor, Docker, Linux, Windows Server

LANGUAGES

Persian (Farsi)	Native
English	Professional Working Proficiency | IELTS Academic: Overall 6.0 - Speaking 6.5, Reading 6.0, Writing 6.0, Listening 6.0